

Abstract

A two-stage framework forecasts returns for **individual S&P 100 stocks** and averages them into sector signals. This phase rebuilds the data foundation — **point-in-time index membership**, a ten-year sample, leak-free feature selection — and **reinstated Ridge**. Once corrected, out-of-sample sector selection is **indistinguishable from chance**.

Prior Work

- **Karatas & Hirsa (2021)**: predict eight sector ETFs, hold the top four.
- **Previous COST group**: move prediction to individual stocks, aggregate to sectors.
- **This phase**: rebuild the universe, add a linear baseline, retest the conclusions.

Data & Universe

- **119 S&P 100 stocks, 9 sectors**, from monthly point-in-time membership (Bloomberg).
- **120 monthly observations**, Feb 2016 – Jan 2026; chronological 77.5 / 22.5 split.
- Macro indicators per sector; **RFE on the train fold**.

Methodology

- 1 Train Ridge / ESN / LSTM / GRU per stock (119 × 4)
- 2 Equal-weight average of eligible constituents → 9 sectors
- 3 Rank sectors → hold Top 4 at 25% each, monthly
- 4 Evaluate vs equal-weight 9-sector benchmark

What Changed This Phase

- **Point-in-time membership** evaluated at the decision date. Universe **58 → 119 stocks**; survivorship bias removed.
- **Sample extended** to 120 months, from roughly five years.
- **Leakage closed**: RFE on the training fold, Sharpe pairing corrected, a duplicated percentage change removed, seeds fixed.
- **Ridge reinstated** after two prior groups dropped it; LSTM cut from 5 h to 1 h per configuration.
- **New diagnostics**: per-sector R^2 , top-four hit rate, constituent-level audit trail.

Key Findings

- **Selection is at chance out of sample**. Top-four hit rate **0.42–0.55** against a **0.506** baseline; in-sample 0.50–0.68.
- **Ridge beats ESN in 20 of 25 grid configurations** — mean Sharpe **1.13 vs 0.87** — while fitting the training period three times worse.
- **Good portfolio statistics are static tilts**. GRU holds the *same four sectors* in all 27 test months.
- **Longer windows only look better**. The test window shrinks from 27 to 17 months as they grow: $r = -0.57$.
- **Predictability is uneven**. Utilities, Industrials, Finance and Energy forecast better than Technology, HealthCare and Consumer Discretionary.

Sector Selection vs Chance

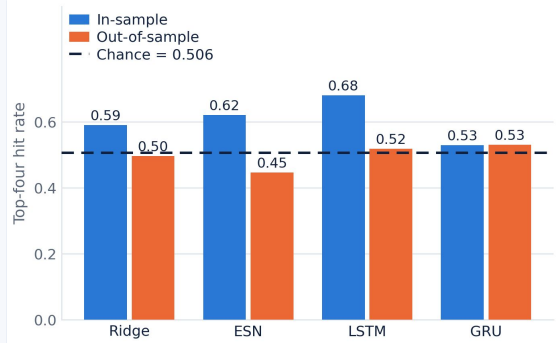


Fig. 1 Mean top-four hit rate over the nine sectors. Every model separates from chance in training and returns to it out of sample.

The Window Artefact

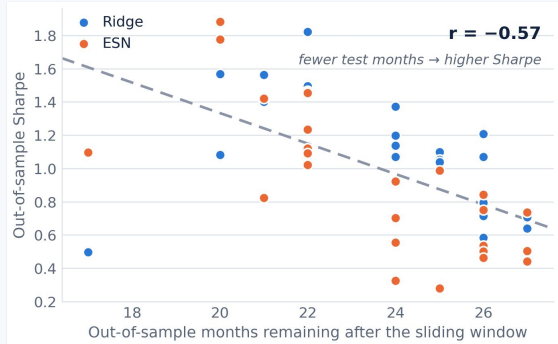


Fig. 2 All 50 grid configurations. Longer windows eat a fixed 120-month sample, so the best scores come from the smallest test sets.

At a Glance

0.42–0.55

Out-of-sample hit rate
(chance 0.506)

34 / 36

Sector-model cells with
negative test R^2

20 / 25

Grid configurations where
Ridge beats ESN

$r = -0.57$

Test Sharpe vs months of
out-of-sample data left

In-Sample vs Out-of-Sample

Diagnostic	In-samp.	Out-of-samp.
Top-four hit rate	0.50–0.68	0.42–0.55
Sector R^2	+0.07 to +0.66	neg. in 34/36
Mean Sharpe · Ridge	0.21	1.13
Mean Sharpe · ESN	0.61	0.87

Conclusion & Future Work

Takeaway: once the universe is point-in-time and a linear baseline is in the comparison, the framework's apparent skill disappears.

FUTURE WORK

1. Fix monthly resampling: it takes the **last daily return**, not the month.
2. Hold the out-of-sample window **fixed** across the grid.
3. Add trivial baselines: buy-and-hold and 12–1 momentum.
4. Pool observations across stocks: ~90 → ~10,700 examples.